

Kristiania University College
PGR211- Advance Programming for Data Science
Worksheet 2

1. Write programs using **for loop** that read a line of input as a string and print
 - a. Only the uppercase letters in the string.
 - b. Every second letter of the string.
 - c. The string, with all vowels replaced by an underscore.
 - d. The number of digits in the string.
 - e. The number of vowels in the string.
2. Write a program using **for loop** that reads a word and prints each character of the word on a separate line. For example, if the user provides the input "Harry", the program prints

```
H
a
r
r
y
```

3. Write a program using **for loop** that reads a word and prints the word in reverse. For example, if the user provides the input "Harry", the program prints

```
yrrah
```

4. Write a program using **nested for loop** to produce the following shape.

```
*
**
***
****
*****
```

5. Write a program using **nested for loop** to draw the following shapes.

```
XXXXX
XX
XXXXX
XX
XX
```

```
XX
XX
XX
XX
XXXXX
```

6. Write a program using **for loop** which prints the positive odd numbers till the user input.

Sample run:

Enter a positive integer: 10

The odd number(s) till number 10 ---> 1 3 5 7 9

7. Write a program using **nested for loops** which prints the sequence of stars reduced by one for each line depending upon the user input.

Sample run:

Enter the number of stars you want in the first row: 6

The star sequence is

```
*****
*****
****
***
**
*
```

8. Write a program using **for loop** which prints the factorial of a positive integer given by user.
[In mathematics, the factorial of a positive integer n , denoted by $n!$, is the product of all positive integers less than or equal to n]. For example:

$$0! = 1$$

$$1! = 1$$

$$2! = 2 \times 1 = 2$$

$$3! = 3 \times 2 \times 1 = 6$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

.

.

$$n! = n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 3 \times 2 \times 1$$

Sample run:

Enter n: 5
5! = 120

Another sample run:

Enter n: 0
0! = 1

9. Write a program using **for loop** and **if/elif/else statements** that gets monthly sales of 5 employees from the user and then calculates and prints the income according to the following table:

Monthly Sales	Income
≥ 30000	575 + 20% of the sale
25000 – 30000 (25000 included)	520 + 10% of the sale
20000 – 25000 (20000 included)	500 + 5% of the sale
< 20000	400 + 2% of the sale

Sample run:

Enter monthly sales: 25000
Income is 3020.0
Enter monthly sales: 20000
Income is 1500.0
Enter monthly sales: 30000
Income is 6575.0
Enter monthly sales: 5000
Income is 500.0
Enter monthly sales: 26000
Income is 3120.0

10. The factorial of a number is the product of all the integers from 1 to that number. For example, the factorial of 6 is $1*2*3*4*5*6 = 720$. Factorial is not defined for negative numbers, and the factorial of zero is one, $0! = 1$. Write a function to find the factorial of a number passed to the function.
11. Write a program using a function called **integerPower** that returns the value of x^y . Assume that y is a positive, nonzero integer and x is an integer. Your function called **integerPower** will get x and y values from the user and return the value of x^y .
12. Write a program using a function called **multiple** that determines for a pair of integers whether the second integer is a multiple of the first. The function should take 2 integer arguments and return appropriate values if the second one is a multiple of the first or not.

13. Write a menu driven program with the following choice of submenu items:

Enter 1 to convert from Celsius to Fahrenheit

Enter 2 to convert from Fahrenheit to Celsius

Select 1 or 2:

- Use a function **fahrenheit_to_celsius** which returns Celsius equivalent of a Fahrenheit temperature.

$$C = (5/9)*(F-32)$$

- Use a function **celsius_to_fahrenheit** which returns Fahrenheit equivalent of a Celsius temperature.

$$F = (9/5)*C + 32$$

Use the functions above to write a program which converts from Celsius to Fahrenheit or from Fahrenheit to Celsius and displays the result. Your program should repeat as long as the user enters 1 or 2 as the choice.

Sample run:

Enter 1 to convert from Celsius to Fahrenheit

Enter 2 to convert from Fahrenheit to Celsius

Select 1 or 2: 1

Enter temperature in Celsius: 5

5.00 Celsius = 41.00 Fahrenheit

Enter 1 to convert from Celsius to Fahrenheit

Enter 2 to convert from Fahrenheit to Celsius

Select 1 or 2: 2

Enter temperature in Fahrenheit: 33

33.00 Fahrenheit = 0.56 Celsius

Enter 1 to convert from Celsius to Fahrenheit

Enter 2 to convert from Fahrenheit to Celsius

Select 1 or 2: -1

STOP!

14. Write a program to compute and display a person's weekly salary as determined by the following specification "If the hours worked are less than or equal to 40, the person receives \$8 per hour; else the person receives \$320 plus \$12 for each hour worked over 40 hours". The program should request the hours worked as input and should display the salary as output. A sample run would be as follows:

```
Please enter the hours worked: 20
Your salary is: 160 Dollars
```

Another sample run:

```
Please enter the hours worked: 60
Your salary is: 560 Dollars
```

15. A senior salesperson is paid \$800 a week and a junior salesperson \$375 a week. Write a program that accepts as input a salesperson's status in the character variable. If the status entered is 's', the senior person's salary will be displayed and else the junior person's salary should be output. A sample run would be as follows:

```
Please enter the status of the salesperson (s/j): s
The salary of the senior salesperson is $800 per week.
```

16. Write a program with first **while loop** and then **for loop** that calculates the value of $f(x)$ for x between 1 and 5 in increments of 1.

$$f(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}$$

A sample run would be as follows:

```
x = 1 --> f(x) = 2.71
x = 2 --> f(x) = 7.00
x = 3 --> f(x) = 16.38
x = 4 --> f(x) = 34.33
x = 5 --> f(x) = 65.38
```

17. Write a program that simulates the toss of a coin. The result can be either head or a tail. Assume that you toss the coin 1000 times. The program should report the number of heads and tails.
18. Write a function to return the volume of a sphere with the input argument for the radius of the sphere. The formula for the volume of the sphere is $\frac{4}{3}\pi r^3$, where r is the radius of the sphere.
19. Write a function, which calculates the value of the following function. Your function should take x as float and return a float value.

$$g(x) = \begin{cases} -x^2 + 4 & \text{if } x < 1 \\ 2x - 1 & \text{if } x \geq 1 \end{cases}$$